

Vincent Jarvis

vjarvis@uwaterloo.ca • linkedin.com/in/vincent-jarvis • vincentjarvis.com

SKILLS

Computer-Aided Design: SolidWorks, Onshape, AutoCAD, Fusion 360, Cura, xTool XCS

Manufacturing: 3D Printing (Bambu Lab, Markforged), Laser cutter, CNC, Woodshop Machines & Tools

Languages: Python, C++, Java

EXPERIENCE

UW Formula Electric

Sep. 2022 - Ongoing

University of Waterloo

Waterloo Ontario

- Directed the **design and optimization of a motor cooling catch can**, conducting extensive research on **5+ material and geometry options** to minimize weight while maximizing internal volume and coolant capacity
- Applied Design for Manufacturing and Assembly (**DFMA**) principles in SolidWorks to create a lightweight, manufacturable design that improved cooling efficiency and simplified assembly, reducing part count by **25%**
- Investigated and benchmarked multiple catch can configurations, analyzing manufacturing methods, wall thickness, and material compatibility to select an **optimized aluminum variant** with superior heat dissipation

First Robotics Competition

Sep. 2024 – June 2025

Ten Ton Academy

West Vancouver, BC

- Led a 9 member** Design and Build subteam, prioritizing efficiency and robustness to earn multiple top 10 finishes, including 6th of 50+ teams, with zero major mechanical failures through **24** matches
- Engineered a modular FRC drivetrain, integrating four Swerve modules with eight **780W** BLDC motors, optimizing the CAD for durability, a heavyweight base, and omnidirectional mobility under competition constraints
- Designed an angled intake funnel in **Onshape** for precise and fast payload alignment into the end effector, incorporating a folding mechanism to allow space for endgame operations while maintaining a **95%** success rate
- Developed all CAD models for a 2 stage cascade pulley elevator lift powered by dual Kraken X60 motors, maximized for structural rigidity and precise alignment, allowing for a **57"** extension in **0.25s**, with a **6.7kg** end effector

Robotics Judge Volunteer

Oct. 2022 - Mar. 2024

Ten Ton Academy

West Vancouver, BC

- Collaborated** with fellow judges and event coordinators to align scoring criteria, streamline judging procedures, and maintain objectivity, resulting in smoother and more transparent evaluations
- Strengthened **leadership, communication**, and evaluation skills by engaging with diverse teams and providing **guidance** under time constraints, preparing for future roles in engineering and team management

PROJECTS

Rocket Car | AutoCAD, Fusion 360, xTool XCS

Jan. 2024

- Modeled a customized baking soda–vinegar powered rocket engine and chassis in **Fusion 360**, applying iterative CAD simulations, resulting in a lightweight frame with a **150%** increase in thrust efficiency
- Created and manufactured laser-cut components in **AutoCAD**, generating toolpaths with xTool XCS and selecting appropriate materials, producing precision parts with improved consistency between iterations

Robotic Hand | Onshape, Cura

Jul. 2023

- Designed and 3D printed a fully functional robotic hand modeled after the **human skeletal structure**, leveraging iterative CAD refinement, resulting in a lifelike demonstration tool for robotics outreach events
- Prototyped finger designs in **Onshape**, evaluating geometry at small scales for load-bearing capacity and material flexibility, which improved finger strength by **40%** while minimizing print failures during fabrication

AWARDS

Design 11 Award | Rockridge Secondary

Sep. 2024

- Recognized for perseverance, innovation, and mastery across nine diverse engineering design projects

Presidential Scholarship of Distinction | University of Waterloo

May. 2025

- Granted for top students achieving a 95%+ admission average, recognizing exceptional academic excellence

EDUCATION

University of Waterloo

Bachelor of Applied Science, Honours Mechanical Engineering

Waterloo, ON

2025-2030